Measuring User Experience

itlinhph

Content

I. User Experience Overview
II. Metric for User Experience
III. Web Vitals & Mobile vitals
IV. Error collect

I. USER EXPERIENCE OVERVIEW

User Experience

User experience (UX) refers to all aspects of user interaction with a product, application, or system.

Factors that Influence UX:

- Useful
- Usable
- Desirable
- Findable
- Accessible
- Credible

User Experience

Design Strategy:

- Focus on beauty: UI
- Focus on business: ROI, net sales revenue,...
- Focus on user experience: What do user need? What problems? Improve user interactive

=> Have to combine all!

Measuring User Experience

Many people seem to think of the user experience as some nebulous quality that can’t be measured.

Some usability metrics:

- How long does it take users to create an order using mobile app/web?
- How many errors do users make when trying to use a feature?
- Task success rates, task times, number of mouse clicks or keystrokes, self-reported ratings,...

Measuring User Experience

- User experience metrics based on large-scale behavioral data
- Most of metrics still largely *business-centered* rather than *user-centered*.
- UX metrics -> *what users do*
- Goals: understand *why they do* it and improve it.

Step to measure User Experience:

- ➔ Determine goals (study goal, user goals)
- ➔ Choosing the right metrics
- ➔ Measure metrics
- ➔ Data analysis & make decision

Measuring User Experience - Determine goals

STUDY GOALS: How the data will ultimately be used within the product development life cycle.

- Focus on identifying ways of making improvements. EX: aspects of the product work well for the users, the most common errors,...
- Focuses on evaluating against a certain set of criteria. Ex: Comparing several products to each other, how does our product compare against the competition?

Measuring User Experience - Determine goals

USER GOALS: knowledge about users and what they are trying to accomplish. Are user use product everyday? Are they likely to use the product only once or just a few times?

- ➔ **Performance**: all about *what the user actually does* in interacting with the product.
  - which users can successfully accomplish a task or set of tasks
  - time takes to perform each task
  - number of mouse clicks
  - number of errors committed
- ➔ **Satisfaction**: all about *what the user says or thinks* about their interaction with the product
  - have opinions about the product
  - report level easy to use, confusing,...

Measuring User Experience - Choose the right metrics

Considering about problems:

- Metrics take too much time to collect
- Metrics cost too much money
- Metrics are not useful when focusing on small improvements
- Metrics don't help us understand causes
- Data are too noisy
- Difficult to collect reliable data with a small sample size
- Metrics are not understood or appreciated by management
- Budgets and timelines

Measuring User Experience - Choose the right metrics

- Metrics take too much time to collect: 6.x, 1.x, 5.x
- Metrics not useful, don't help us understand causes: 1.x, 2.x
- Budgets and timelines

Measuring User Experience - Choose the right metrics

Usability Study Scenario	Task Success	Task Time	Errors	Efficiency	Learn-ability	Issues-Based Metrics	Self-Reported Metrics	Behavioral and Physiological Metrics	Combined and Comparative Metrics	Live Website Metrics	Card-Sorting Data
1. Completing a transaction	X			X		X	X			X	
2. Comparing products	X			X			X		X		
3. Evaluating frequent use of the same product	X	X		X	X		X				
4. Evaluating navigation and/or information architecture	X		X	X							X
5. Increasing awareness							X	X		X	
6. Problem discovery						X	X				
7. Maximizing usability for a critical product	X		X	X							
8. Creating an overall positive user experience							X	X			
9. Evaluating the impact of subtle changes										X	
10. Comparing alternative designs	X	X				X	X		X		

II. METRICS FOR UX

Metric for UX

- Performance Metrics
- Issue based
- Self-reported metrics
- Behavioral and physiological metrics
- Conversion Rate

Metrics for UX - Performance Metrics

Rely not only on user behaviors but also on the use of scenarios or tasks:

1. **Task success:** measures how effectively users are able to complete a given set of tasks:
   - Binary success
   - Level success
2. **Time-on-task:** measures how much time is required to complete a task.
3. **Errors**: reflect the mistakes made during a task
4. **Efficiency**: The amount of effort a user expends to complete a task, ex: number of clicks in a website to complete buy a product.
5. **Learnability**: measure how performance changes over time. It examine how and when participants reach proficiency in using a product.

Metrics for UX - Issue based

UX issues define:

- Anything that prevents task completion
- Anything that creates some level of confusion
- Anything that produces an error
- Not seeing something that should be noticed
- Assuming something is correct when it is not
- Assuming a task is complete when it is not
- Performing the wrong action
- Misinterpreting some piece of content
- Not understanding the navigation

Metrics for UX - Issue based

Collect metrics:

- **In-Person**: Reporting what they are doing, what they are trying to accomplish
- **Automated**: Task success, time, ease-of-use rating, and verbatim comments
- **Log:** system log report issued.

Severity ratings based on UX:

- Impact on the user experience
- Predicted frequency of occurrence
- Impact on the business goals

	Few users experiencing a problem	Many users experiencing a problem
Small impact on the user experience	Low severity	Medium severity
Large impact on the user experience	Medium severity	High severity

Severity of issue

Metrics for UX - Self-reported metrics

Ask users to tell you about their experience with it:

- Forms
- Rating scales
- Lists of attributes
- Open-ended questions: List the top three things you liked the most about this application.

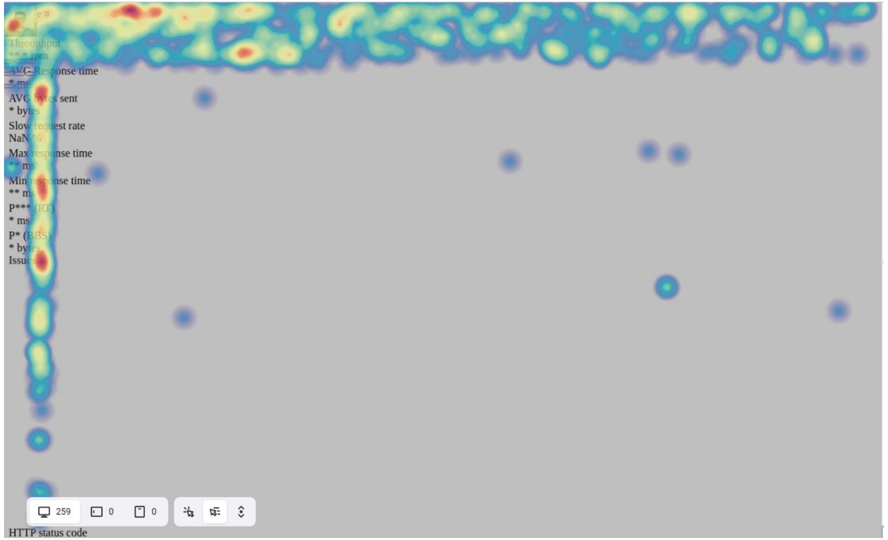
Metrics for UX - Behavioral and physiological metrics

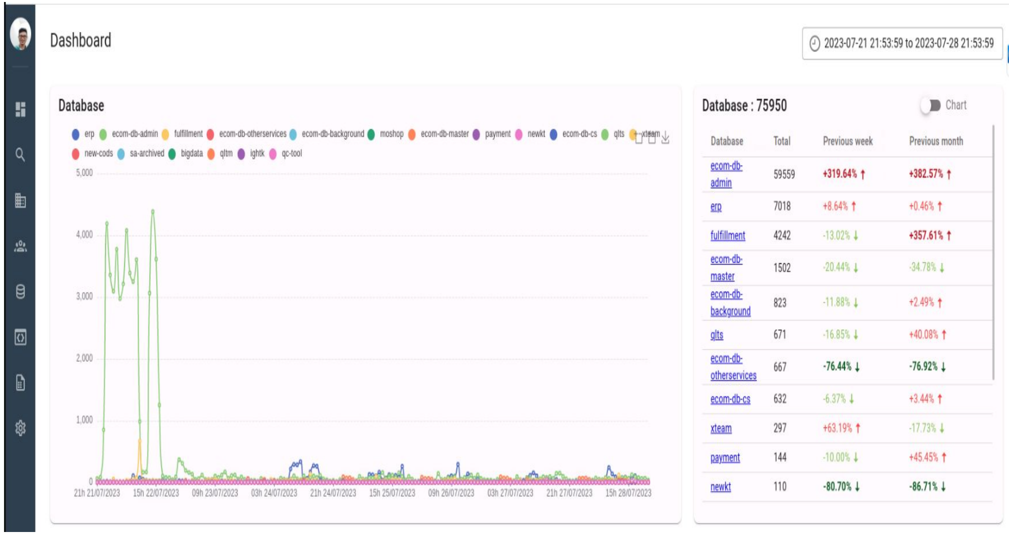
Observing behavior:

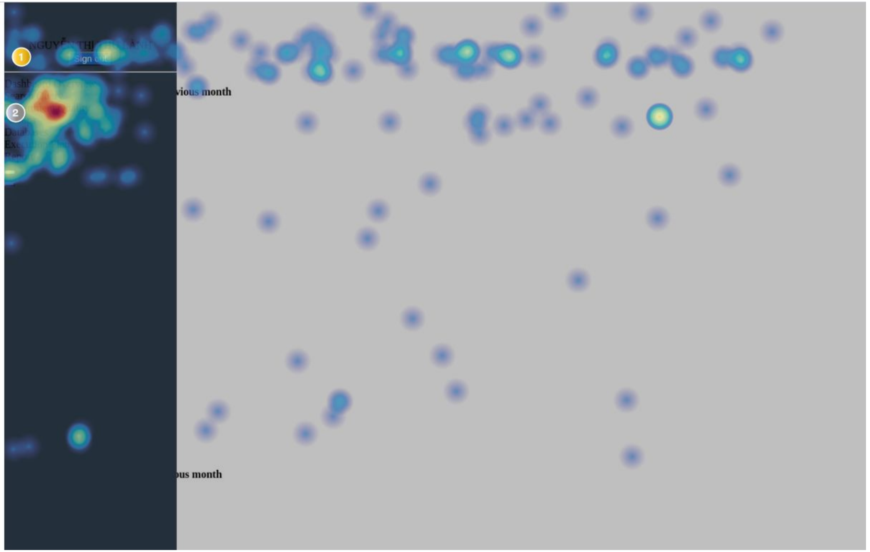
- Comments good/bad about products

Requiring equipment to capture:

- Facial Expressions
- Eye-Tracking
- Mouse event, click map, heart map


AVG Response time
AVG bytes sent
Slow request rate
Max response time
Min response time
HTTP status code


Dashboard
2023-07-21 21:53:59 to 2023-07-28 21:53:59
Database
Database : 75950
Chart
Database Total Previous week Previous month
ecom-db-admin 59559 +319.64% +382.57%
erp 7018 +8.64% +0.46%
fulfillment 4242 -13.02% +357.61%
ecom-db-master 1502 -20.44% -34.78%
ecom-db-background 823 -11.88% +2.49%
qlts 671 -16.85% +40.08%
ecom-db-otherservices 667 -76.44% -76.92%
ecom-db-cs 632 -6.37% +3.44%
xteam 297 +63.19% -17.73%
payment 144 -10.00% +45.45%
newkt 110 -80.70% -86.71%




Phan Hồng Lĩnh
SIGN OUT
Dashboard Overview
Search Query
Department Quality
Groups
Databases
Execution Plan
Report
Setting
Dashboard
2023-07-21 21:53:59 to 2023-07-28 21:53:59
Database
Database : 75952
Chart
Database
Total
Previous week
Previous month
ecom-db-admin 59561 +319.65% +382.59%
erp 7018 +8.64% +0.46%
fulfillment 4242 -13.02% +357.61%
ecom-db-master 1502 -20.44% -34.78%
ecom-db-background 823 -11.88% +2.49%
qlts 671 -16.85% +40.08%
ecom-db-otherservices 667 -76.44% -76.92%
Group : 75952
Chart
Group
ERP 51577 +534.64% +673.15%
OS 14613 +9.72% -8.71%

Metrics for UX - Conversion Rate

Conversion Rate: Number of tasks/users that reach a goal.

Ex: Number of user who create an order when go to create order page.

A/B Testing: compares the performance of two versions of content to see which one appeals more to visitors/viewers

UX metrics framework for web application

PULSE METRICS:

- **P**age views
- **U**ptime
- **L**atency
- **SE**ven-day active users

HEART METRICS:

- **H**appiness: aspects of user experience, like satisfaction, visual appeal, likelihood to recommend)
- **E**ngagement: user’s level of involvement with a product.
- **A**doption: how many new users start using a product during a given time period.
- **R**etention: how many of the users from a period are still present in some later time period
- **T**ask success: Efficiency (time to complete a task), effectiveness (percent of tasks completed), error rate,...

Metrics for UX - Summary

- Determine goals first, then choose the right metrics
- Two main aspects of the user experience: *performance* (what user does) & *satisfaction* (what user think or feel)
- Budgets and timelines need to be planned out well in advance when running any usability studies involving metrics
- With small numbers of users -> issue-based metric
- For summative usability studies: Recommend from 50-100 users
- If possible, compare our product to others, we may track additional metrics from the standard set
- For web system:
    - **HEART** metrics are more user-centric than **PULSE**
    - Raw counts will go up and need to be normalized; ratios, percentages, or averages per user are often more useful.
    - Metrics based on web logs: need filter out traffic from automated sources (crawlers, spammers, ajax, frontend automatic,...)

III. Web vitals & Mobile vitals

Web vitals - Overview

- **Web Vitals** is an initiative by Google to provide unified guidance for quality signals that are essential to delivering a great user experience on the web.
- Google has provided a number of tools over the years to measure and report on performance.
- The Web Vitals initiative aims to simplify the landscape, and help sites focus on the metrics that matter most, the **Core Web Vitals**.

Web vitals

Types of metrics:

- **Perceived load speed**: how quickly a page can load and render all of its visual elements to the screen.
- **Load responsiveness**: how quickly a page can load and execute any required JavaScript code in order for components to respond quickly to user interaction
- **Runtime responsiveness**: after page load, how quickly can the page respond to user interaction.
- **Visual stability**: do elements on the page shift in ways that users don't expect and potentially interfere with their interactions?
- **Smoothness**: do transitions and animations render at a consistent frame rate and flow fluidly from one state to the next?

Web vitals

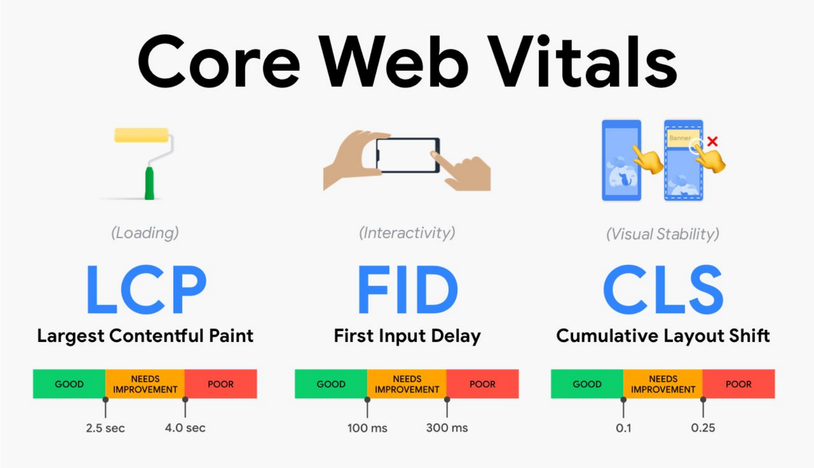
Important metrics to measure:

- First Contentful Paint (FCP)
- Largest Contentful Paint (LCP)
- First Input Delay (FID)
- Interaction to Next Paint (INP)
- Time to Interactive (TTI)
- Total Blocking Time (TBT)
- Cumulative Layout Shift (CLS)
- Time to First Byte (TTFB)

Core web vitals

Core Web Vitals represents a distinct facet of the user experience, measurable, reflects the real-world experience of a critical user-centric outcome.

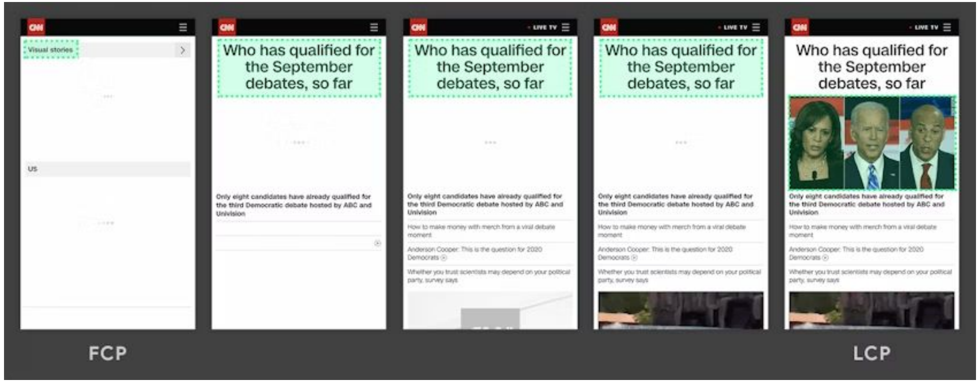
- *Largest Contentful Paint (LCP)*
- *First Input Delay (FID)*
- *Cumulative Layout Shift (CLS)*



[9 most effective Core Web Vitals optimizations for 2023]

Core web vitals

Largest Contentful Paint (LCP): measures loading performance. To provide a good user experience, LCP should occur within 2.5 seconds of when the page first starts loading.



LCP can change during the time

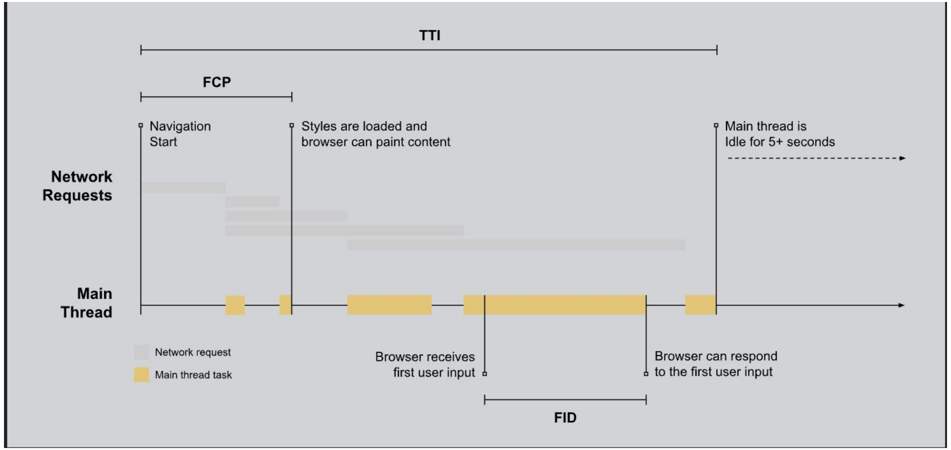
LCP

Largest Contentful Paint

Core web vitals

First Input Delay (FID): measures interactivity. To provide a good user experience, pages should have a FID of 100 milliseconds or less.

Focuses on response from discrete actions like clicks, taps, and key presses.



Core web vitals

Cumulative Layout Shift (CLS): measures visual stability, the largest burst of layout shift scores for every unexpected layout shift that occurs during the entire lifespan of a page.

To provide a good user experience, pages should maintain a CLS of 0.1 or less.

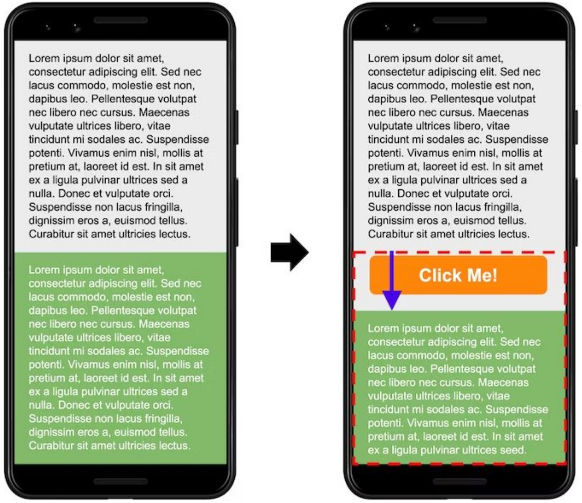
```
CLS = impact fraction * distance fraction
```


Core web vitals

[Cumulative Layout Shift (CLS)](): measures visual stability, the largest burst of layout shift scores for every unexpected layout shift that occurs during the entire lifespan of a page.

To provide a good user experience, pages should maintain a CLS of 0.1 or less.

```
CLS = impact fraction * distance fraction
```



CLS = 0.5*0.14 = 0.07

CLS

Cumulative Layout Shift

Others web vitals

First Contentful Paint (FCP): user-centric metric for measures the time from when the page starts loading to when any part of the page's content is rendered on the screen.

Time to First Byte (TTFB) is a metric that measures the time between the request for a resource and when the first byte of a response begins to arrive.

Web vitals - Tool measure

- Chrome User Experience Report
- PageSpeed Insights
- Search Console (Core Web Vitals report)
- Web-vitals JavaScript library

Core Web Vitals

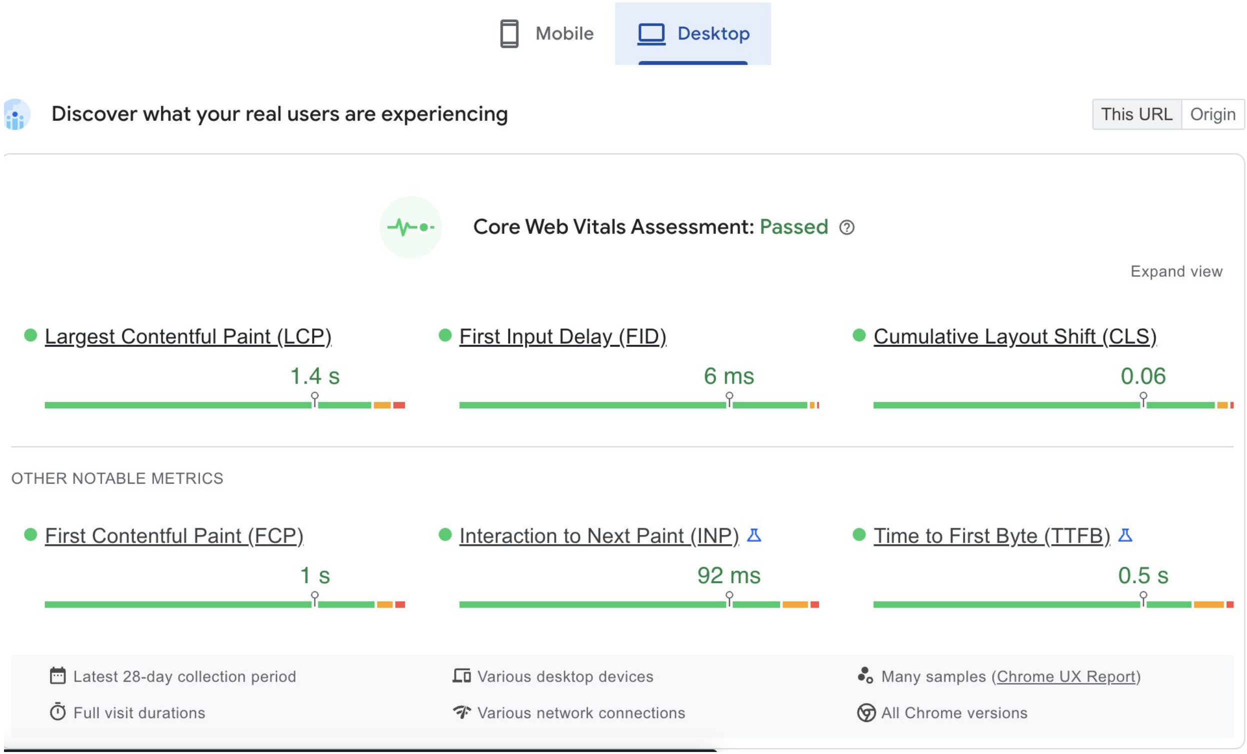
Now in your favorite developer tools

	LCP	FID	CLS
PageSpeed Insights	✅	✅	✅
Chrome UX Report Brand new API, BigQuery and Dashboard	✅	✅	✅
Search Console	✅	✅	✅
Chrome DevTools	✅	TBT	✅
Lighthouse	✅	TBT	✅
Web Vitals Extension	✅	✅	✅

LCP = Largest Contentful Paint, FID = First Input Delay, CLS = Cumulative Layout Shift, TBT = Total Blocking Time

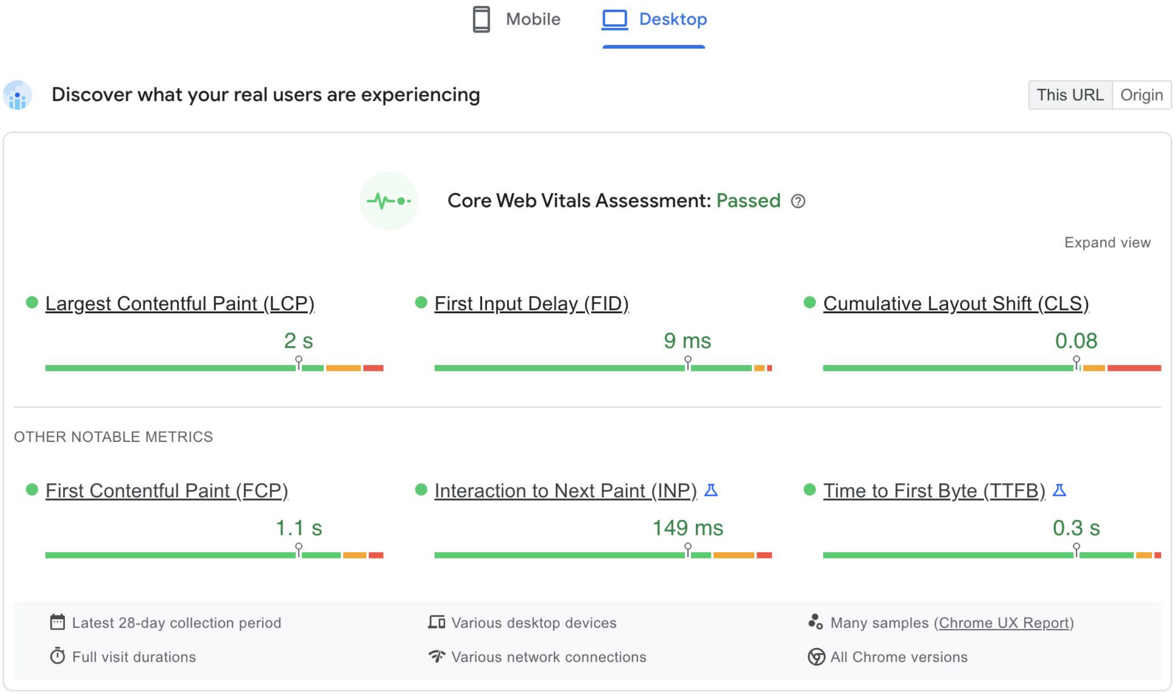
Web vitals samples

Lazada



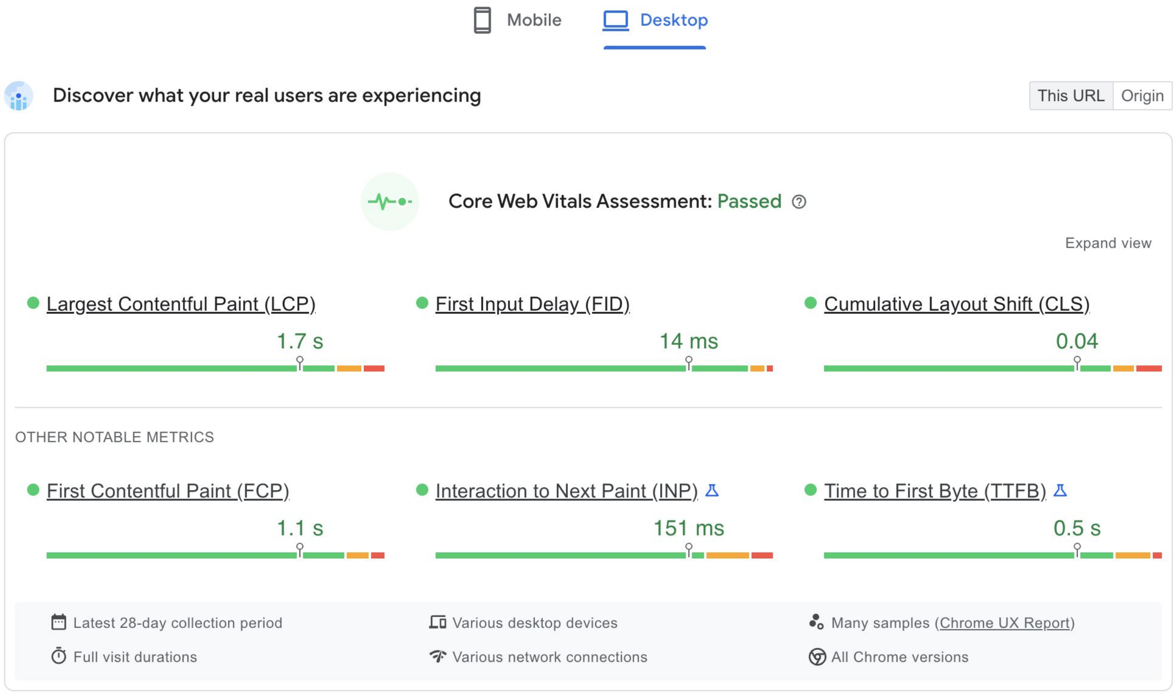
Web vitals samples

Tiki



Web vitals samples

Amazon



Android Vitals/Mobile vitals

Android vitals is an initiative by *Google* to improve the technical quality of Google Play apps on Android devices.

When an opted-in user runs your app, their Android device logs information about aspects of quality including stability metrics, performance metrics, battery usage, and permission denials.

This data is aggregated by default in Google Play:

- Google Play Console in the Android vitals dashboard
- Google Play Developer Reporting API.

Core Android Vitals

Two core vitals:

- user-perceived crash rate
- user-perceived ANR rate

BAD BEHAVIOR THRESHOLD To maximize your title's visibility on Google Play, please keep it below these thresholds.		
	Overall (average across devices)	Per phone model
User-perceived crash rate	1.09%	8%
User-perceived ANR rate	0.47%	8%

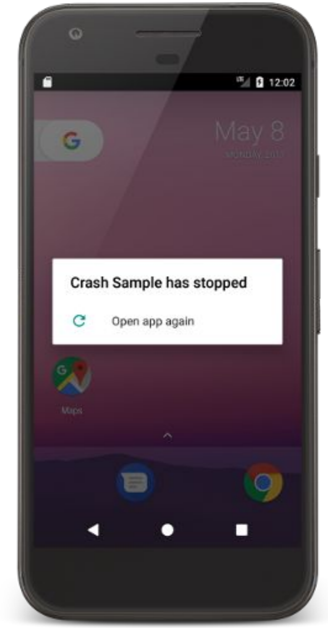
Core vitals - Crash rate

Crash rate:

- Crash rate: The percentage of your daily active users who experienced any type of crash.

Bad behavior:

- Overall bad behavior threshold: At least **1.09%** of daily active users experience a user-perceived crash, across all device models.
- Per-device bad behavior threshold: At least **8%** of daily active users experience a user-perceived crash, for a single device model.



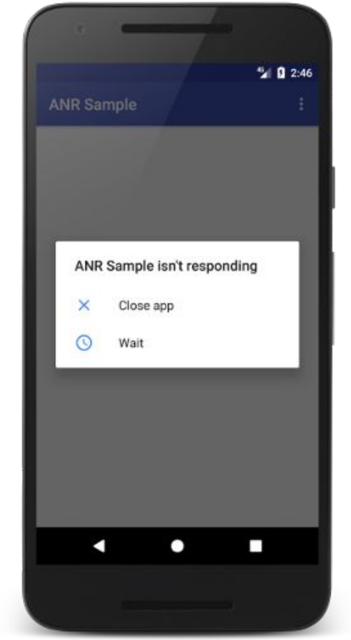
Core vitals - ARN

ANR: Application Not Responding

ANR rate: The percentage of your daily active users who experienced any type of ANR.

Bad behavior:

- Overall bad behavior threshold: At least **0.47%** of daily active users experience a user-perceived ANR, across all device models.
- Per-device bad behavior threshold: At least **8%** of daily users experience a user-perceived ANR, for a single device model.



Other vitals

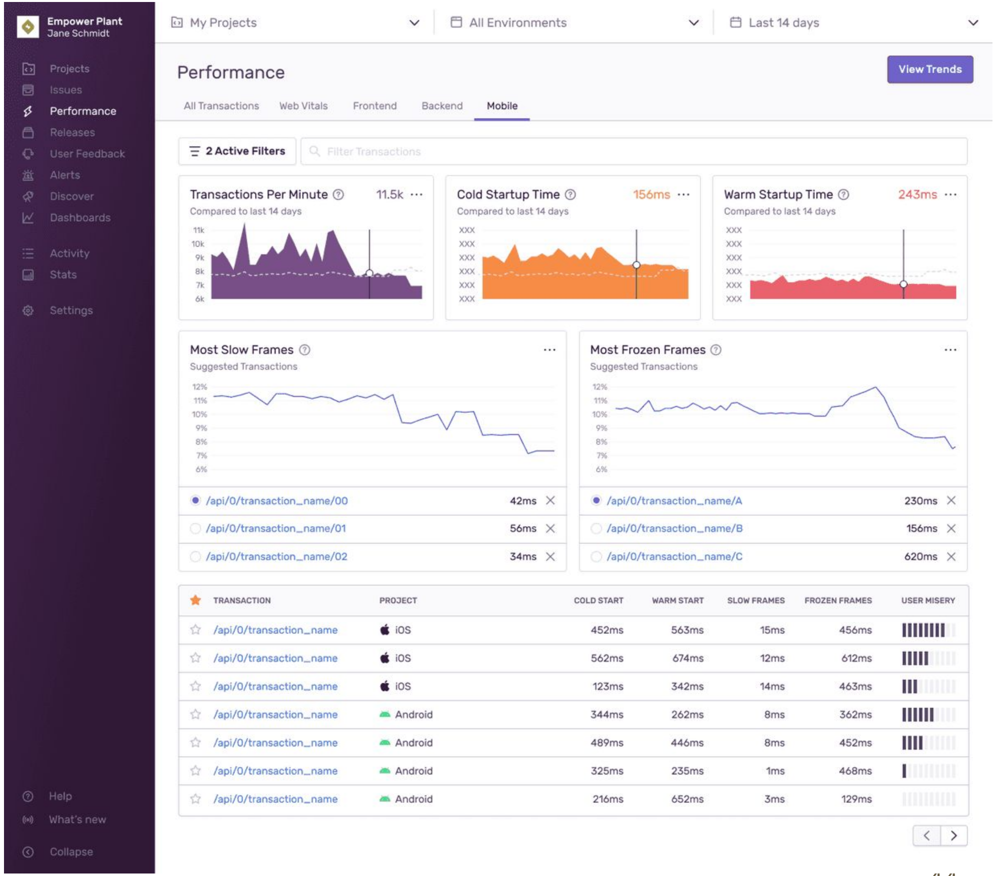
Other vitals:

- Slow rendering
- Frozen frames
- Excessive wakeups
- Stuck partial wake locks
- Excessive background Wi-Fi scans
- Excessive background network usage
- App startup time
- Permission denials

Sentry mobile vitals

App Start:

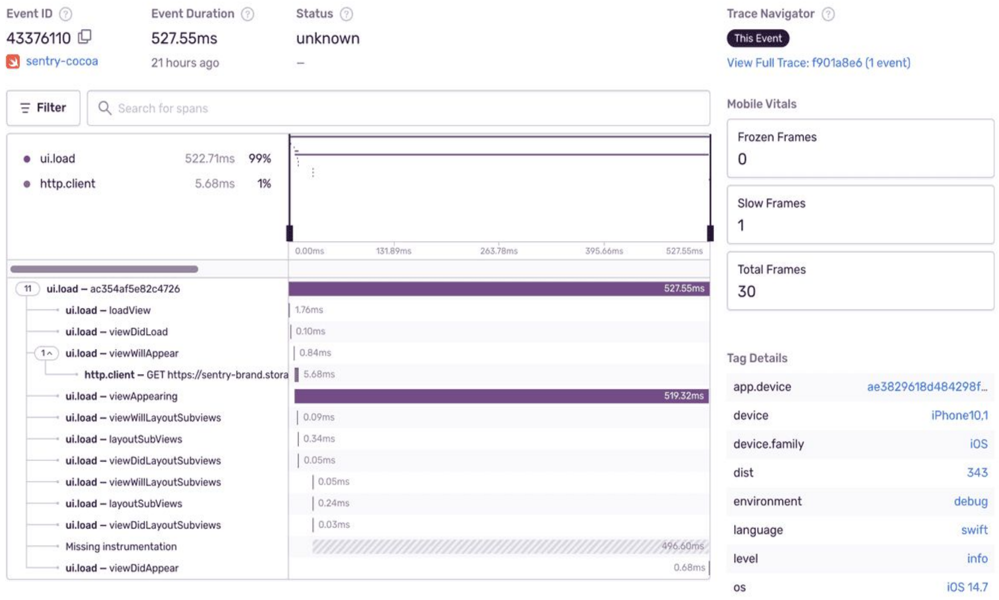
- Measures cold starts and warm starts.
- On iOS, Apple recommends app take at most 400ms to render the first frame
- On Android, the Google Play console warns when a cold start takes longer than 5 seconds or a warm start longer than 2 seconds.



Sentry mobile vitals

Slow and Frozen Frames:

- Slow Frames: Using 60 fps, slow frames are frames that take more than 16 ms (Android) or 16.67 ms (iOS) to render.
- Frozen Frames: Frozen frames are frames that take longer than 700 ms to render.



Mobile vitals - Tools

- Firebase Performance Monitoring
- JankStats library
- Sentry mobile vitals

V. Error collect (System error)

Error collect

A kind of issue-based metric but related to service-centric

Source:

- Log file:
  - Database (slow query, error log, metric performance, deadlock,...)
  - System error log
- Web access log
- Debug log
- Sentry event
- App error log (firebase)

Error log collect

Phase 1: Sentry error log

- Integration with Sentry: Web backend, frontend, mobile app
- Collect metric from sentry database

Phase 2: Others system log

- Collect system error log (from infra)
- 4xx, 5xx error
- MySQL: slow query (done), performance schema, sys, information schema, error log file

Summary

- There are a various metrics of UX measure. Base on goals -> Choose a strategy and right metrics.
- Two main aspects of the user experience: *performance* (what user does) & *satisfaction* (what user think or feel)
- Web vitals & mobile vitals is a set of useful metrics for User Experience.
- Google analytics for basic user experience (PULSE metrics, user performance metrics)
- Sentry for Web vitals, Mobile vitals, System Performance metric, Error log
- Access log/system log, issue-based metric for detect incidents
- Deep metrics if we have big budgets & timelines.

References

1. Measuring the User Experience Collecting, Analyzing, and Presenting Usability Metrics (Tom Tullis Bill Albert)
2. Practical Web Analytics for User Experience (Michael Beasley)
3. Measuring the User Experience on a Large Scale: User-Centered Metrics for Web Applications (Kerry Rodden, Hilary Hutchinson, and Xin Fu)
4. Web vitals
5. Android vitals
6. Sentry Architecture